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Time-Series Modelling and Forecasting of Daily Copper Prices

An ARMA Model with a GARCH Extension

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Abstract

Copper is the cornerstone of the Zambian economy, consistently accounting for the bulk of national export earnings, so the statistical behaviour of its price is of direct fiscal and commercial interest. This study models 6,298 daily copper prices (US\$/tonne) spanning Dec 2000–May 2025. The price level is non-stationary, so the analysis works with daily log-returns. Following the Box–Jenkins methodology, the autocorrelation and partial-autocorrelation functions together with the Bayesian Information Criterion identify an **AR(2)** model for the conditional mean. The fitted ARMA(2,0) removes the linear dependence in returns, but its residuals display strong volatility clustering and heavy tails, and out of sample it does not outperform a naive random walk. The study therefore extends the specification to an **ARMA(2,0)–GARCH(1,1)** model with Student- t innovations. The extension lowers the BIC from 24,742 to 21,872, eliminates the autoregressive conditional heteroskedasticity from the standardised residuals, and captures both the persistence of volatility ($\alpha_1 + \beta_1 = 0.974$) and the fat tails (t d.f. ≈ 4.6) characteristic of commodity returns. The results imply that copper prices are close to weak-form efficient in the mean, while their *risk* is highly predictable — the practically useful object to forecast.

Contents

Abstract	1
1 Introduction	4
1.1 Background	4
1.2 Statement of the Problem	4
1.3 Aim and Objectives	4
1.4 Scope and Limitations	4
1.5 Organisation of the Study	5
2 Data and Methodology	5
2.1 Data	5
2.2 Returns and Stationarity	5
2.3 The ARMA Model	5
2.4 Diagnostics and Out-of-Sample Evaluation	6
2.5 The GARCH Extension	6
2.6 Software	6
3 Results: The ARMA Model	6
3.1 Price Behaviour and Returns	6
3.2 Stationarity Tests	7
3.3 Order Selection: ACF, PACF and BIC	8
3.4 The Fitted ARMA(2,0) Model	8
3.5 Residual Diagnostics	9
3.6 Out-of-Sample Forecast Accuracy	10
4 The Improved Model: ARMA–GARCH	11
4.1 Motivation	11
4.2 The Fitted ARMA(2,0)–GARCH(1,1) Model	11
4.3 Model Comparison	12
4.4 What the GARCH Model Reveals	12
4.5 Why It Is an Improvement	13
5 Conclusions and Recommendations	13
5.1 Summary of Findings	13
5.2 Conclusions	14

5.3	Recommendations	14
5.4	Areas for Further Research	14
References		14
A Full R Output		16
A.1	Stationarity tests	16
A.2	Fitted ARMA(2,0)	16
A.3	ARMA(2,0)-GARCH(1,1)-t: variance equation and diagnostics	16

1 Introduction

1.1 Background

Copper is the single most important commodity in the Zambian economy, routinely contributing the majority of the country’s foreign-exchange earnings and a substantial share of government revenue (World Bank, 2023). Because the national budget, the exchange rate and the mining sector’s investment decisions are all sensitive to the copper price, understanding and forecasting that price is of considerable practical importance.

Commodity prices such as copper are typically studied as financial time series. The seminal framework is that of Box and Jenkins (1976), who formalised the autoregressive moving-average (ARMA) class of models and an iterative *identification–estimation–diagnostic-checking* procedure for fitting them. A recurring empirical finding for prices of metals, equities and currencies is that, although the *level* of a price is non-stationary and behaves much like a random walk, its *volatility* is highly predictable: large changes tend to be followed by large changes and small by small. This “volatility clustering” motivated the autoregressive conditional heteroskedasticity (ARCH) model of Engle (1982) and its generalisation, the GARCH model of Bollerslev (1986).

1.2 Statement of the Problem

A classical ARMA model assumes that the error variance is constant over time (homoskedasticity) and, for interval forecasts, that the errors are normally distributed. Daily commodity returns routinely violate both assumptions. If an analyst fits an ARMA model to copper returns and stops there, the point estimates may be reasonable but the standard errors, confidence intervals and any value-at-risk figures derived from the model will be unreliable, because the model has ignored the time-varying risk that dominates the data. This study asks two questions: *(i)* what ARMA model best describes the conditional mean of daily copper returns, and *(ii)* does explicitly modelling the conditional variance with a GARCH component materially improve the model?

1.3 Aim and Objectives

The aim is to build, validate and improve a time-series model for daily copper prices. The specific objectives are:

1. to transform the price series to a stationary form suitable for ARMA modelling;
2. to identify and estimate an ARMA model for the conditional mean using the ACF, PACF and the Bayesian Information Criterion (BIC);
3. to assess the adequacy of that model through residual diagnostics and out-of-sample forecast accuracy; and
4. to extend the model with a GARCH component and quantify the resulting improvement.

1.4 Scope and Limitations

The study is univariate: it uses only the history of the copper price itself and does not incorporate macroeconomic or market covariates such as the US dollar index, Chinese industrial demand or inventory levels. The data are daily spot prices in US dollars per tonne; exchange-rate effects on the kwacha value of copper are outside the scope. The analysis is conducted entirely in R, ensuring full reproducibility.

1.5 Organisation of the Study

Section 2 describes the data and the methodology. Section 3 presents the ARMA modelling and its diagnostics — the original analysis. Section 4 develops the GARCH extension and explains why it constitutes an improvement. Section 5 concludes. Full R output is reproduced in the appendix, and the complete code and data are available in the project repository.

2 Data and Methodology

2.1 Data

The dataset comprises 6,299 daily observations of the copper price in US dollars per tonne. The magnitude and trajectory of the series (rising from roughly US\$1,800/tonne in 2000 to a peak above US\$10,800/tonne) are consistent with London Metal Exchange cash-settlement quotations. Three data-quality issues were corrected before analysis: 36 dates had been keyed with the year 19xx instead of 20xx (the entire January–February 2022 block), which were shifted forward by one century; 1 duplicate date was removed; and the series was sorted chronologically. The cleaned series contains 6,298 observations spanning 01 December 2000 to 30 May 2025.

2.2 Returns and Stationarity

Let P_t denote the closing price on day t . The price level trends strongly and is non-stationary, so the analysis uses the continuously-compounded (log) return, expressed in percent,

$$r_t = 100 \times \ln\left(\frac{P_t}{P_{t-1}}\right). \quad (1)$$

Stationarity is assessed with the Augmented Dickey–Fuller (ADF) test of Dickey and Fuller (1979), whose null hypothesis is a unit root, and the KPSS test of Kwiatkowski et al. (1992), whose null is stationarity.

2.3 The ARMA Model

An ARMA(p, q) model for the return is

$$r_t = \mu + \sum_{i=1}^p \phi_i r_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}, \quad \varepsilon_t \sim \text{i.i.d.}(0, \sigma^2). \quad (2)$$

Candidate orders are read from the sample autocorrelation function (ACF) and partial autocorrelation function (PACF): an AR(p) process produces a PACF that cuts off after lag p , and an MA(q) process an ACF that cuts off after lag q (Box & Jenkins, 1976). Among competing specifications the preferred model minimises the Bayesian Information Criterion, $\text{BIC} = -2\ell + k \ln n$ (Schwarz, 1978), where ℓ is the maximised log-likelihood, k the number of parameters and n the sample size. The BIC is reported alongside Akaike’s AIC (Akaike, 1974); BIC is preferred here because it penalises complexity more heavily and is consistent for the true order in large samples.

2.4 Diagnostics and Out-of-Sample Evaluation

An adequate model should leave white-noise residuals. This is tested with the Ljung–Box Q -statistic (Ljung & Box, 1978) on the residuals and on the squared residuals, the latter detecting ARCH effects; the ARCH-LM test of Engle (1982) provides a second check, and the Jarque–Bera test (Jarque & Bera, 1987) examines normality. Forecast performance is judged out of sample: the model is estimated on all but the final 252 trading days and used to produce one-step-ahead forecasts over that hold-out window, which are compared against a naive random-walk benchmark ($\hat{P}_t = P_{t-1}$) using the root-mean-square error (RMSE) and mean absolute error (MAE).

2.5 The GARCH Extension

To capture volatility clustering, the constant-variance assumption is relaxed by letting the conditional variance σ_t^2 follow a GARCH(1,1) process (Bollerslev, 1986):

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad \omega > 0, \alpha_1, \beta_1 \geq 0, \alpha_1 + \beta_1 < 1, \quad (3)$$

with the AR(2) mean retained from Section 3. The quantity $\alpha_1 + \beta_1$ measures the persistence of volatility shocks. To accommodate the heavy tails of commodity returns, the innovations are assumed to follow a standardised Student- t distribution, whose degrees-of-freedom parameter is estimated jointly with the model.

2.6 Software

All computation uses R: `readxl` for data import, `tseries` for the unit-root and normality tests, `forecast` for ARMA estimation and forecasting, `lmtest` and `FinTS` for diagnostics, and `rugarch` for the GARCH model.

3 Results: The ARMA Model

3.1 Price Behaviour and Returns

Figure 1 plots the price level and its logarithm. The series wanders without reverting to a fixed mean — the visual signature of non-stationarity — with the 2008 commodity crash, the 2011 and 2021–2024 booms all clearly visible. Figure 2 shows the log-returns, which fluctuate around zero but in distinct calm and turbulent episodes: the hallmark of volatility clustering that motivates Section 4.

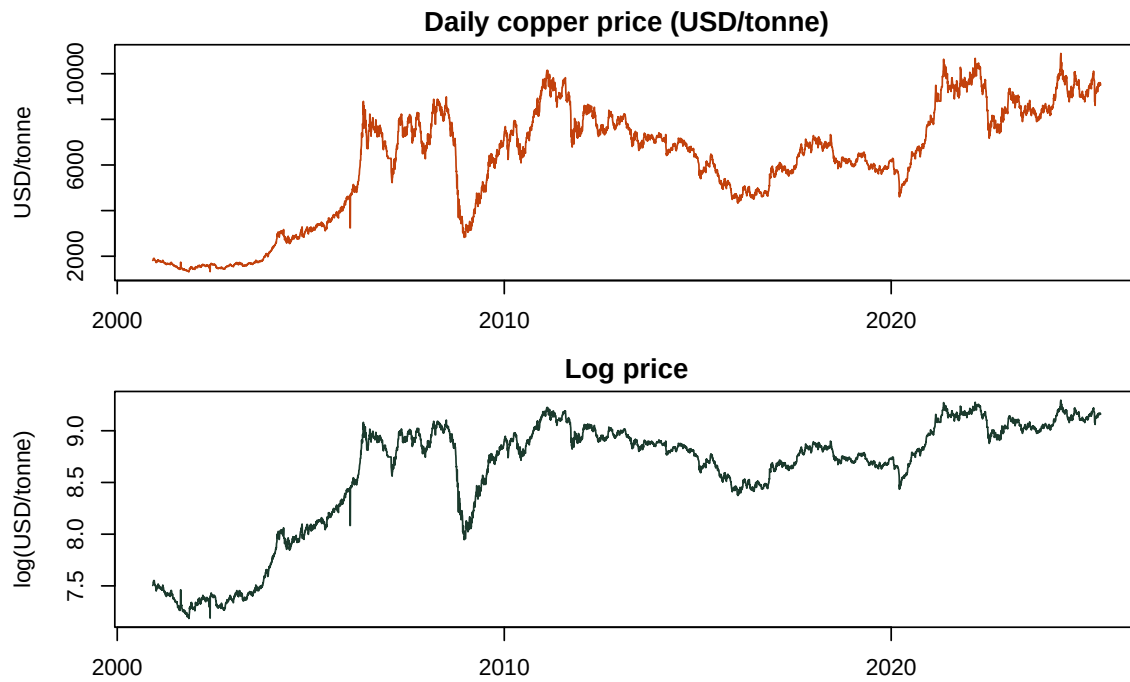


Figure 1: Daily copper price (top) and log price (bottom). The wandering, non-mean-reverting level is characteristic of a non-stationary, near-random-walk series.

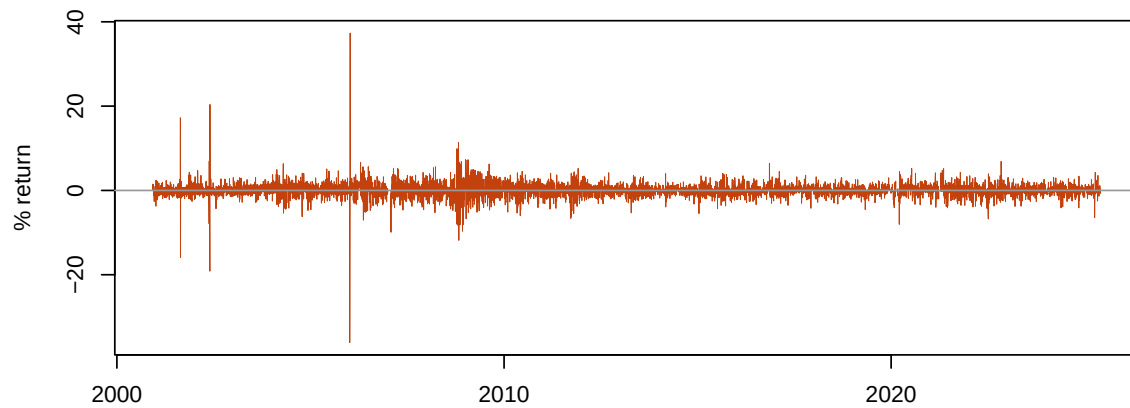


Figure 2: Daily log-returns (%). Volatility arrives in clusters — calm stretches punctuated by turbulent ones — which a constant-variance ARMA model cannot represent.

3.2 Stationarity Tests

The tests confirm the visual impression. On the price level the ADF test fails to reject a unit root ($p = 0.377$) while KPSS rejects stationarity ($p \leq 0.01$). On the returns the verdict reverses: ADF strongly rejects a unit root ($p \leq 0.01$) and KPSS does not reject stationarity ($p \geq 0.1$). The returns are therefore stationary and suitable for ARMA modelling; equivalently, the log price is integrated of order one, $I(1)$.

3.3 Order Selection: ACF, PACF and BIC

The ACF and PACF of the returns (Figure 3) both show significant spikes at lags 1 and 2 that then die away — a pattern consistent with a low-order autoregression. The BIC grid search over $p, q \in \{0, \dots, 5\}$ confirms this: the minimum BIC is attained by the AR(2) specification, ARMA(2,0). The eight best models are listed below (R output).

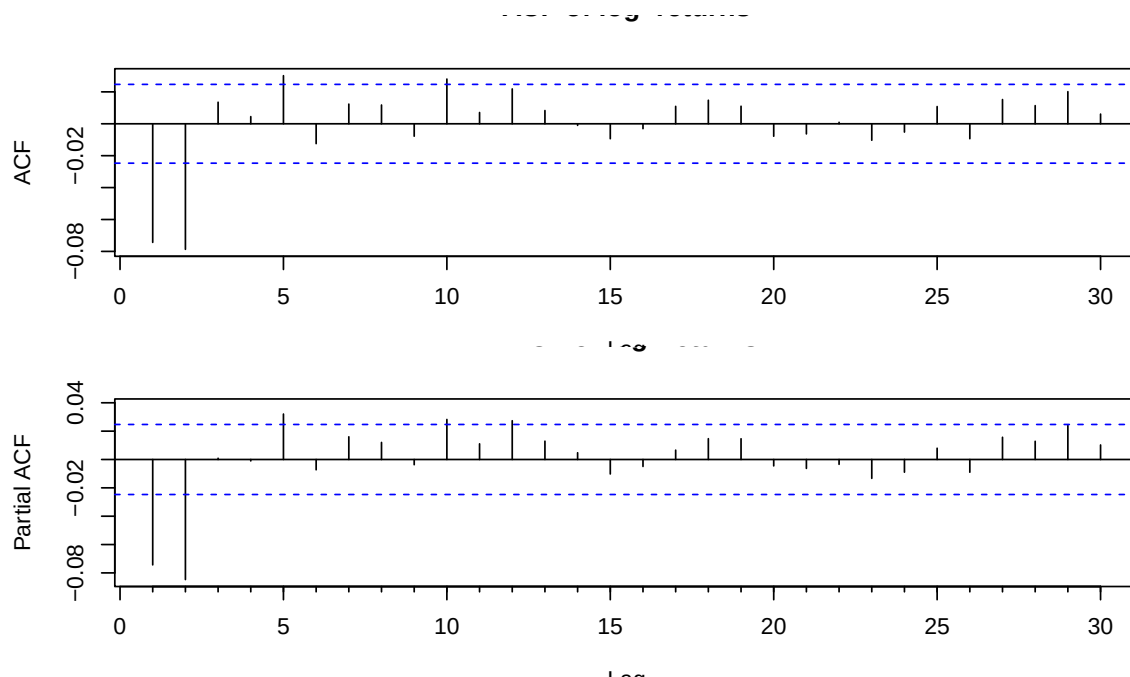


Figure 3: ACF and PACF of daily log-returns. Significant spikes at lags 1–2 in both functions point to a low-order AR process.

```
head(gres[order(gres$BIC), ], 8) # candidate ARMA orders ranked by BIC (lowest = best)
##   p q   AIC   BIC
## 3  2 0 24715 24742
## 13 0 2 24717 24744
## 18 0 3 24717 24751
## 14 1 2 24717 24751
## 9  2 1 24717 24751
## 4  3 0 24717 24751
## 22 0 4 24718 24759
## 5  4 0 24719 24760
```

3.4 The Fitted ARMA(2,0) Model

Estimating the selected model gives the coefficients below. Both autoregressive terms are highly significant ($p < 10^{-9}$); the constant (mean daily return) is not significantly different from zero, consistent with prices that drift only weakly.

```
fit <- Arima(ret, order = c(2,0,0), include.mean = TRUE)
coeftest(fit)
##
## z test of coefficients:
##
```

```
##           Estimate Std. Error z value Pr(>|z|)
## ar1        -0.0806     0.0126  -6.42 1.3e-10 ***
## ar2        -0.0847     0.0126  -6.74 1.5e-11 ***
## intercept    0.0263     0.0186   1.41  0.16
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3.5 Residual Diagnostics

The model captures the *linear* dependence in returns: the Ljung–Box test on the residuals is insignificant at the 5% level ($Q_{20} p = 0.097$), so no autocorrelation remains in the mean. However, two assumptions fail decisively. The Ljung–Box test on the *squared* residuals and the ARCH-LM test both reject the null of no conditional heteroskedasticity at any conventional level ($p < 10^{-10}$), and the Jarque–Bera test rejects normality. In words, the ARMA residuals are uncorrelated but *not* independent — their variance is strongly predictable and their distribution is heavy-tailed (Figure 4).

```
Box.test(residuals(fit), lag = 20, type = "Ljung-Box", fitdf = 2) # autocorrelation in mean
##
## Box-Ljung test
##
## data: residuals(fit)
## X-squared = 26.1, df = 18, p-value = 0.097
Box.test(residuals(fit)^2, lag = 20, type = "Ljung-Box") # ARCH effects (squared)
##
## Box-Ljung test
##
## data: residuals(fit)^2
## X-squared = 1222, df = 20, p-value <2e-16
ArchTest(residuals(fit), lags = 12) # Engle's ARCH-LM test
##
## ARCH LM-test; Null hypothesis: no ARCH effects
##
## data: residuals(fit)
## Chi-squared = 1534, df = 12, p-value <2e-16
```

Table 1: Out-of-sample one-step forecast accuracy over the final 252 trading days (price level, USD/tonne). Lower is better.

Model	RMSE	MAE
ARMA(2,0)	118.33	86.17
Random walk	117.68	84.87

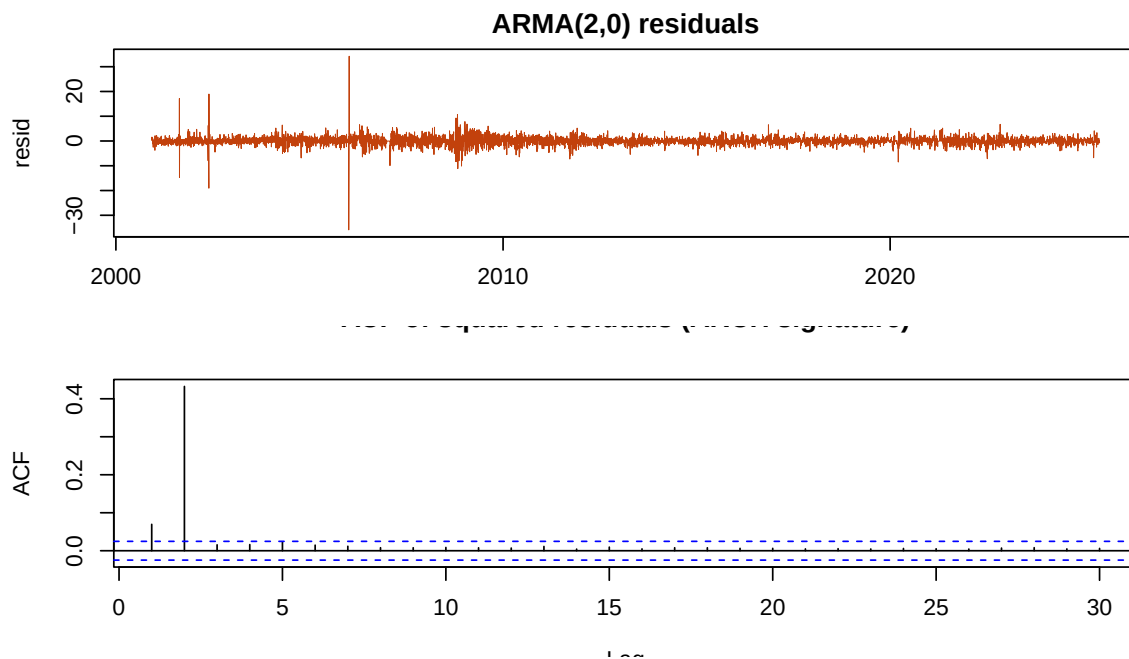


Figure 4: ARMA(2,0) residuals. The squared-residual ACF (bottom) shows large, slowly-decaying autocorrelation — undeniable ARCH that the homoskedastic model ignores.

3.6 Out-of-Sample Forecast Accuracy

The model was re-estimated on all but the final 252 trading days and used to produce one-step-ahead forecasts over that window. The point-forecast accuracy is essentially identical to — and marginally worse than — a naive random walk, and the direction of the next day’s move is called correctly only 49.2% of the time (no better than a coin toss). This is the expected result for a near-efficient price: the small but statistically significant AR(2) structure is *not* economically exploitable for point forecasting.

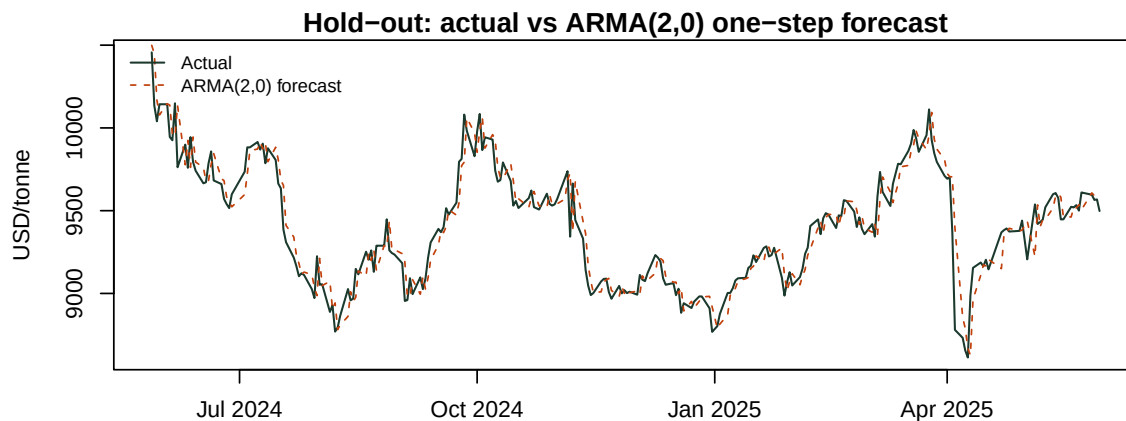


Figure 5: Actual vs. one-step ARMA forecast over the hold-out window. The forecast tracks the lagged price so closely that it is visually indistinguishable from a random walk.

4 The Improved Model: ARMA–GARCH

4.1 Motivation

Section 3.5 established that the ARMA(2,0) residuals violate the constant-variance and normality assumptions: there is pronounced volatility clustering and the tails are far heavier than Gaussian. Ignoring this has real consequences. The standard errors in the ARMA output assume homoskedasticity, so the reported significance and any confidence or value-at-risk intervals are distorted; and a Gaussian interval forecast will be far too wide in calm periods and far too narrow in turbulent ones. The remedy is to model the conditional variance explicitly. The AR(2) mean is retained and a GARCH(1,1) variance with Student- t innovations is added.

4.2 The Fitted ARMA(2,0)–GARCH(1,1) Model

The estimated parameters are shown below (R output from `rugarch`). All variance parameters are highly significant. Two features stand out. First, volatility is highly persistent: $\alpha_1 + \beta_1 = 0.974$, close to one, meaning shocks to volatility decay only slowly. Second, the estimated Student- t degrees of freedom is ≈ 4.6 — well below the ~ 30 at which the t becomes effectively normal — formally confirming heavy tails. Notably, once the variance is modelled the second AR term shrinks toward insignificance, indicating that part of the apparent mean-dependence in Section 3 was in fact unmodelled heteroskedasticity.

```
spec <- ugarchspec(variance.model = list(model = "sGARCH", garchOrder = c(1,1)),
  mean.model      = list(armaOrder = c(2,0), include.mean = TRUE),
  distribution.model = "std")           # Student-t innovations
fg <- ugarchfit(spec, data = ret, solver = "hybrid")
round(fg$fit$matcoef, 6)
##      Estimate Std. Error t value Pr(>|t|)
## mu      0.023655   0.013756  1.71959 0.085507
## ar1     -0.061497   0.012515 -4.91374 0.000001
## ar2     -0.001257   0.012415 -0.10123 0.919370
## omega    0.079965   0.017305  4.62095 0.000004
## alpha1   0.105385   0.013485  7.81497 0.000000
## beta1    0.868396   0.016901 51.38124 0.000000
```

Table 2: ARMA vs. ARMA-GARCH. The GARCH model wins decisively on fit (logLik, AIC, BIC) and is the only one whose (standardised) residuals pass the no-ARCH tests.

Model	k	logLik	AIC	BIC	LB($\hat{\sigma}_t^2$) p	ARCH-LM p
ARMA(2,0)	4	-12354	24715	24742	0	2.2e-321
ARMA(2,0)-GARCH(1,1)-t	7	-10906	21825	21872	1	1

```
## shape 4.575965 0.284070 16.10861 0.000000
```

4.3 Model Comparison

Table 2 places the two models side by side. The GARCH extension raises the log-likelihood from -12,354 to -10,906 and, despite its three extra parameters, lowers the BIC by almost 2,900 points — an overwhelming margin (a BIC difference above 10 is regarded as decisive). Crucially, the diagnostic columns flip from rejection to non-rejection: the ARCH effects and squared-residual autocorrelation that contaminated the ARMA residuals are absent from the GARCH *standardised* residuals.

4.4 What the GARCH Model Reveals

Figure 6 plots the estimated conditional volatility σ_t . Far from being constant (the flat line the ARMA model implicitly assumes), daily volatility ranges from under 1% in calm periods to over 4% during the 2008 crisis and the 2020 and 2022 shocks. This time-varying risk is the single most important feature of the data, and it is invisible to the ARMA model.

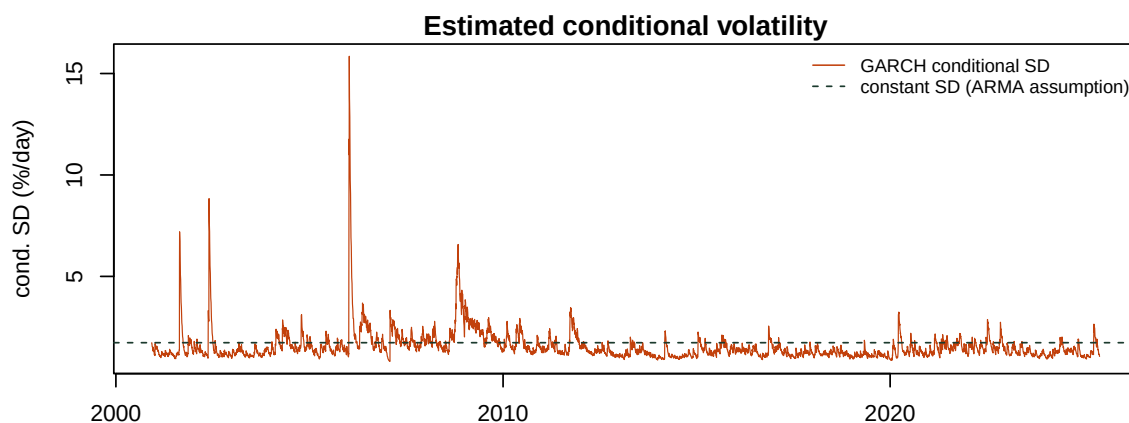


Figure 6: GARCH(1,1) conditional volatility (annualised-scale daily %). Risk is anything but constant; the ARMA model assumes the flat dashed line.

Figure 7 confirms the model is now adequate. The ACF of the squared standardised residuals is flat — the ARCH has been absorbed — and the QQ-plot against the Student- t shows the heavy tails are well described once the right innovation distribution is used.

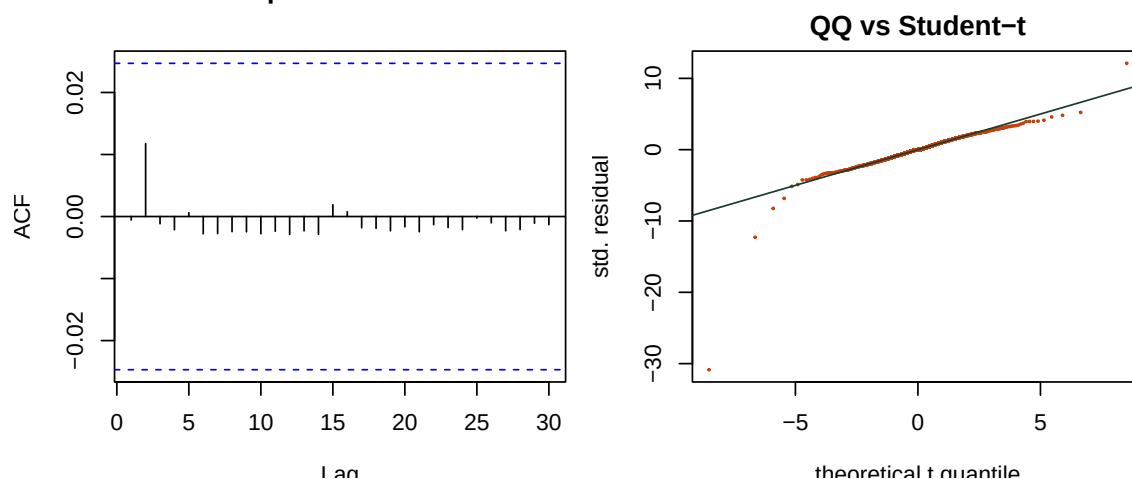


Figure 7: Standardised-residual diagnostics. Left: squared standardised residuals show no remaining ARCH (contrast Figure 4). Right: QQ-plot against the fitted Student- t .

4.5 Why It Is an Improvement

The GARCH extension improves on the pure ARMA model in four concrete ways.

1. **Valid inference.** By modelling the heteroskedasticity rather than ignoring it, the parameter standard errors become reliable, so statements about which terms matter are trustworthy.
2. **Honest, time-varying uncertainty.** The ARMA model attaches the *same* forecast interval to every day; the GARCH model widens its intervals in turbulent regimes and narrows them in calm ones, matching how copper risk actually behaves.
3. **Better density and risk forecasts.** Because it captures volatility clustering and fat tails, the model produces accurate value-at-risk and other tail-risk measures — exactly what a treasury or mining-finance desk needs — where the Gaussian ARMA would understate the probability of large losses.
4. **Decisive statistical superiority.** It is preferred by both AIC and BIC by a wide margin and is the only specification whose residuals pass every diagnostic.

A caveat preserves the honesty of the analysis: the GARCH layer does *not* make the copper price more predictable *in the mean* — the point forecast remains close to a random walk, as Section 3.6 showed. What it makes predictable, and predictable well, is the *risk*. For copper, that is the quantity worth forecasting.

5 Conclusions and Recommendations

5.1 Summary of Findings

Daily copper prices over 2000–2025 are non-stationary and behave, in their level, much like a random walk. The log-returns are stationary, and the ACF, PACF and BIC jointly select an AR(2) model for the conditional mean. That model is statistically valid in the mean but inadequate overall: its residuals exhibit strong volatility clustering and heavy tails, and out of sample it does not beat a naive random walk. Extending it to an ARMA(2,0)–GARCH(1,1) model with Student- t innovations resolves these defects, with a far higher likelihood, decisively lower information criteria, clean standardised residuals, volatility persistence of 0.974 and tail thickness corresponding to about 4.6 degrees of freedom.

5.2 Conclusions

The copper price is approximately weak-form efficient in the mean: past prices carry little exploitable information about the *direction* of future prices. Its variance, by contrast, is strongly and persistently predictable. Any serious model of copper-price behaviour must therefore model the conditional variance; an ARMA model alone is incomplete.

5.3 Recommendations

For analysts and risk managers, copper-price risk should be modelled with a GARCH-type specification and a heavy-tailed innovation distribution, and value-at-risk should be computed from the conditional rather than the unconditional variance. For forecasting the price level itself, a random walk (optionally with drift) is a hard-to-beat benchmark, so resources are better spent on volatility and scenario analysis than on chasing point forecasts. For policymakers reliant on copper revenue, the persistence of volatility implies that price shocks are protracted, strengthening the case for stabilisation buffers.

5.4 Areas for Further Research

Natural extensions include asymmetric volatility models (EGARCH, GJR-GARCH) to test whether price drops raise volatility more than equivalent rises; multivariate or ARIMAX models that bring in the US dollar, Chinese demand and inventory data; regime-switching models for the distinct boom and crash phases visible in Figure 1; and longer-horizon density forecasts for budgeting applications.

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A Full R Output

This appendix reproduces the key console output verbatim for reference and replication.

A.1 Stationarity tests

```
adf.test(price)      # level: H0 = unit root (non-stationary)
##
## Augmented Dickey-Fuller Test
##
## data: price
## Dickey-Fuller = -2.48, Lag order = 18, p-value = 0.38
## alternative hypothesis: stationary
adf.test(ret)        # returns
##
## Augmented Dickey-Fuller Test
##
## data: ret
## Dickey-Fuller = -16.8, Lag order = 18, p-value = 0.01
## alternative hypothesis: stationary
kpss.test(ret)       # returns: H0 = stationary
##
## KPSS Test for Level Stationarity
##
## data: ret
## KPSS Level = 0.148, Truncation lag parameter = 11, p-value = 0.1
```

A.2 Fitted ARMA(2,0)

```
fit
## Series: ret
## ARIMA(2,0,0) with non-zero mean
##
## Coefficients:
##      ar1      ar2    mean
##    -0.081 -0.085  0.026
## s.e.  0.013  0.013  0.019
##
## sigma^2 = 2.96: log likelihood = -12354
## AIC=24715  AICc=24715  BIC=24742
```

A.3 ARMA(2,0)-GARCH(1,1)-t: variance equation and diagnostics

```
infocriteria(fg)      # per-observation IC (rugarch)
##
## Akaike      3.4660
## Bayes      3.4735
## Shibata     3.4660
## Hannan-Quinn 3.4686
cat(sprintf("Volatility persistence a1+b1 = %.4f\n", pers))
```

```
## Volatility persistence a1+b1 = 0.9738
cat(sprintf("Student-t d.f. (shape)      = %.2f\n", tdf))
## Student-t d.f. (shape)      = 4.58
Box.test(z, 20, type="Ljung-Box")          # std resid: autocorrelation
##
## Box-Ljung test
##
## data:  z
## X-squared = 24.4, df = 20, p-value = 0.22
Box.test(z^2, 20, type="Ljung-Box")        # std resid^2: remaining ARCH
##
## Box-Ljung test
##
## data:  z^2
## X-squared = 1.43, df = 20, p-value = 1
ArchTest(z, 12)                            # std resid: ARCH-LM
##
## ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  z
## Chi-squared = 1.21, df = 12, p-value = 1
```

Code and data: <https://github.com/CCheelo/copper-forecasting>